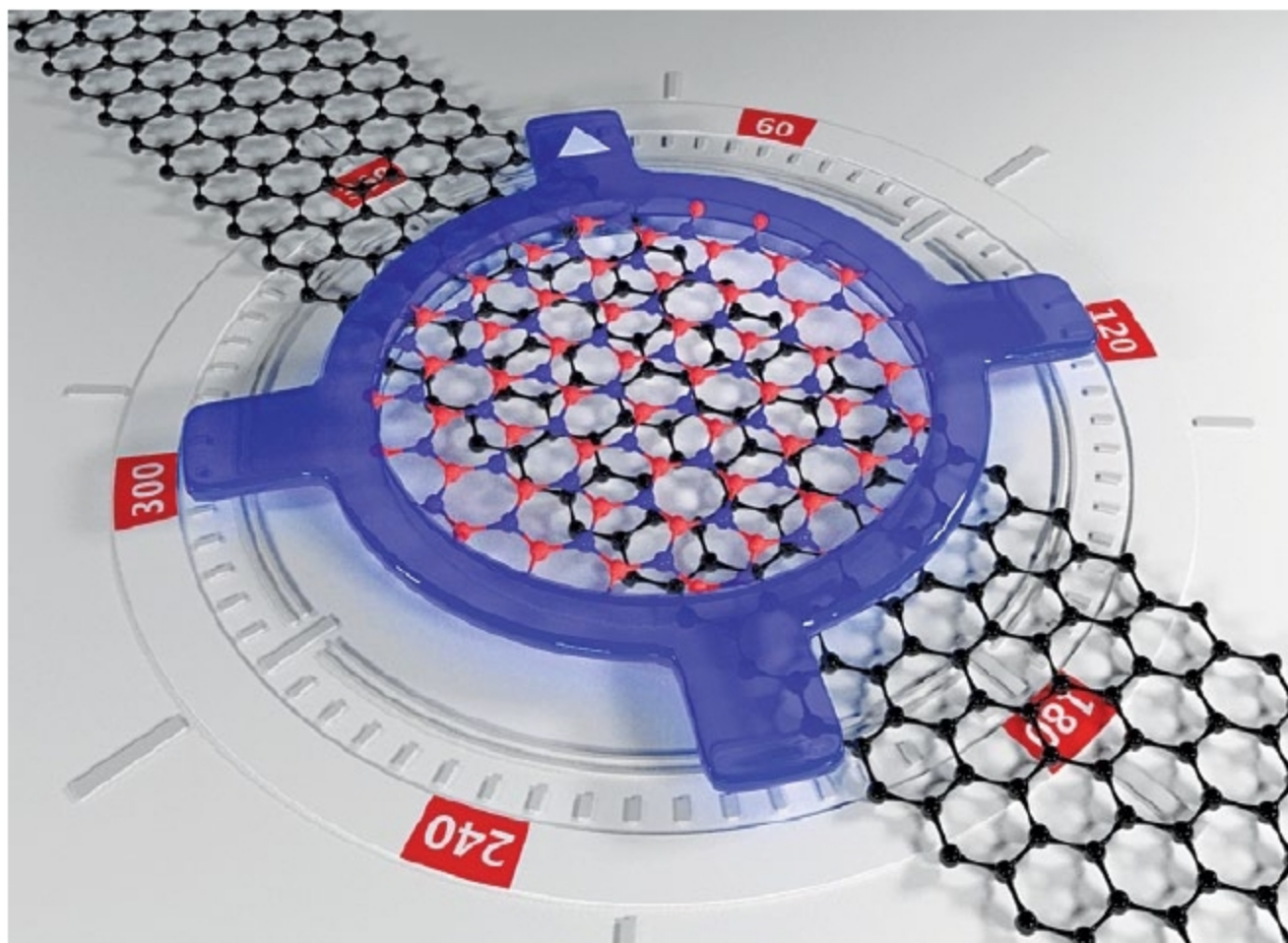




How Twisting Changes Properties Of 2D Materials



Twisted electronics (Credit: <https://physicsworld.com>)

The ability to tune properties of a layered material by varying the twist angle provides the possibility for a single material to perform a variety of functions instead of using many materials for different functions



Dr S.S. Verma

is professor at Department of Physics, Sant Longowal Institute of Engineering and Technology, Sangrur, Punjab

Tunability of a material's properties has always been at the core of material research and development, and scientists are exploring options to find ways of tuning the properties to desired levels, including by doping, stretching, compressing and now twisting. Twisting single-atom layers of graphene and other substances has already shown interesting properties in the past. As an example, here electrons no longer jump from one atom to the next, but tend to flow over the surface of the sheet, called plasmonics.

Plasmonics is being considered in various ways; one of this being the very low resistance involved in flow of electrons and in the interaction happening

in the tera-hertz range (optoelectronics). Researchers are always looking at emerging properties when two layers of graphene and other substances, each one a single atom thick, are layered on top of one another at different angles.

In this direction, twistronics is the study of how the angle (twist) between layers of 2D materials can change their electrical, optical and mechanical properties. Such materials as graphene have shown to have vastly different electrical properties (ranging from non-conductive to super-conductive) depending on angle between the layers.

With the magic angle of twist, scientists have reported evidence of a new kind of quantum optical behaviour in similar